

Bcho NAM Player

Standalone Manual



Standalone | Version 1.7.5

Complete reference for the standalone application on Windows, macOS and Linux. Every control, window, file and message is documented here. For the VST3 / AU plug-in see [PLUGIN_MANUAL.en.md](#).

Contents

1. What Bcho NAM Player is 2. Installing and first launch 3. Map of the window 4. Quick start: sound in five steps 5. Audio Setup: device, latency and routing 6. Signal flow 7. The rack: blocks, order and the eye 8. BLOCK NAM 1 and BLOCK NAM 2 9. Cabinet impulse responses 10. Amplifier controls, knob by knob 11. The eight effects in full 12. MONO and STEREO: two complete rigs 13. Linking effects between L and R 14. Tuner 15. Input Cali (automatic input calibration) 16. TONE3000 inside the player 17. Portable .bnpp presets 18. What is remembered, and where 19. Settings, skins, updates and support 20. Meters and animated cones 21. Mouse and keyboard reference 22. Troubleshooting 23. Technical specifications 24. Integrity, third-party code and licences

1. What Bcho NAM Player is

Bcho NAM Player is a portable standalone guitar processor. It plays Neural Amp Modeler captures (.nam), cabinet impulse responses (.wav) and a reorderable chain of studio effects, and it owns the audio device itself: driver, channels, sample rate, buffer size and physical output routing are configured inside the application.

The player has **two complete and independent signal paths**, left and right. Each path owns its own BLOCK NAM 1, BLOCK NAM 2, cabinet IR, effects rack, block order, gate, tone stack, master level, IR blend, IR cabinet volume, input and output gain, power switch, calibration and tuner. In **MONO** only the left path is audible and the right-hand controls are hidden; in **STEREO** both run at once.

What is in the package

Item	Notes
Bcho NAM Player.exe (or .app / AppImage)	The whole application; no installer, no runtime DLLs
Models	Two demonstration NAM captures. They are not loaded automatically
IRs	The folder the IR browser opens by default. No cabinet IR is bundled
NAM_A2_HEAD.xml	Appears after the first normal exit: your saved session
tone3000.session	Appears only after connecting a TONE3000 account (encrypted)

The window opens centred at its native 1537 x 1023 size whenever the screen allows it. The whole design is responsive - furniture, knobs, lists, meters and text scale together - and cannot be made smaller than 50 % of native size, so no control or legend can ever leave its plate.

2. Installing and first launch

Download the package for your system from the [latest public release](#), extract it, and keep every file and folder together.

System	How to run it
Windows x64	Run Bcho NAM Player.exe. Install your interface manufacturer's ASIO driver for the lowest latency
macOS Apple Silicon	Use the arm64 ZIP, open Bcho NAM Player.app
macOS Intel	Use the x86_64 ZIP, open Bcho NAM Player.app
Linux x86_64	Run the AppImage; <code>chmod +x BchoNAMPlayer - *.AppImage</code> if it will not start

macOS builds are ad-hoc signed but **not** notarized. If Gatekeeper blocks the first launch, right-click the application and choose **Open**.

What the first launch looks like. The player starts in MONO, with every block disabled, no model and no IR loaded, and the audio device set to the system default. A completely bypassed chain passes your dry signal through - it does not mute. Your first job is Audio Setup and then loading a capture.

Every later launch restores the last session you closed normally: controls, both paths, resources, block order, effects and bypass states.

3. Map of the window



The standalone in MONO with a capture, a cabinet and two effects running

The window has three pieces of furniture, from top to bottom:

- 1 - The rack strip.** Preset bay on the left, the reorderable processing blocks in the middle, the tuner button and display, the tuning selector, and on the right the MONO / STEREO switch and the settings gear.
- 2 - The amplifier head.** Input VU, INPUT GAIN and INPUT CALI on the left; the seven main knobs and, above them, the NAM and IR selectors; the two file browsers (NAM models on the left, IR files on the right) with the IR VOL fader between them and the output column; Output VU, OUTPUT GAIN and the red POWER switch on the right.
- 3 - The speaker cabinet.** The Bcho logo and two cones that move with the real output level.



Preset bay, input-mode switch and the first rack blocks

The caption under **LOAD PRESET** always names what the shared controls are editing right now - for example **PATH R • NAM 2**, or **MONO • NAM 1**.

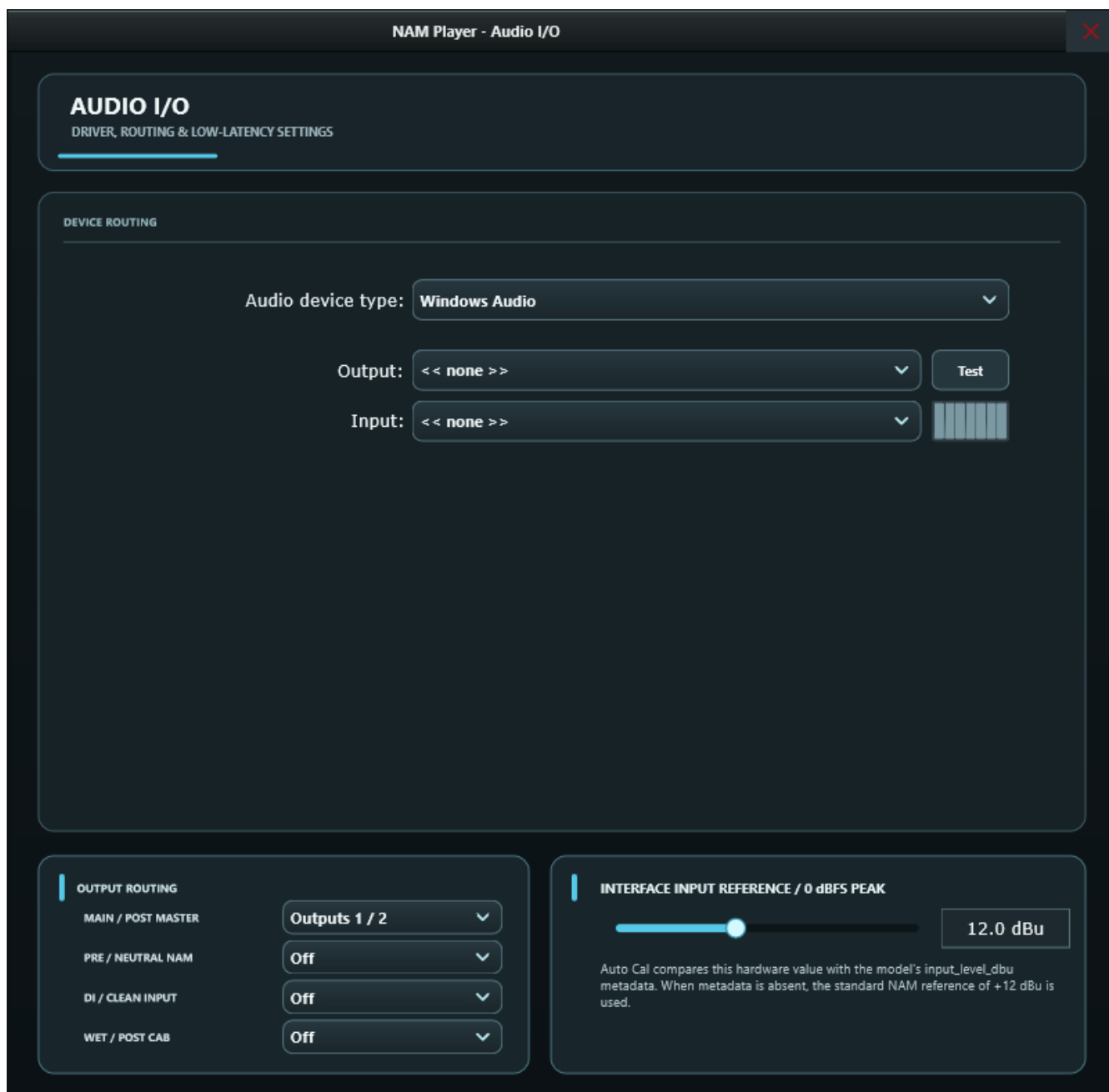
4. Quick start: sound in five steps

1. **Open Audio Setup.** Press the gear (top right of the rack) → **AUDIO SETUP**. Choose your driver and interface, pick the input your guitar is plugged into, set the sample rate and the buffer size, and make sure **MAIN / POST MASTER** points at the outputs your monitors use. 2. **Load an amp capture.** Press **BROWSE LOCAL** under **NAM MODELS** and pick a .nam file or a whole folder. The capture loads, **BLOCK NAM 2** lights up, and the file appears in the list. 3. **Load a cabinet.** Press **BROWSE IR** and pick a .wav impulse response. The IR block lights up. (Skip this if your capture already includes a cabinet - many "amp_cab" captures do.) 4. **Set levels.** Play and watch **INPUT VU**: aim for the green range with the loudest strumming just touching yellow. Use **INPUT GAIN** to trim. Then set **MASTER VOL** and **OUTPUT GAIN** for a comfortable monitoring level. 5. **Play.** Add effects by clicking their blocks; double-click a block to open its editor.

Start at a low monitoring level and avoid hearing your guitar twice - through the interface's own direct monitoring **and** through the player.

5. Audio Setup: device, latency and routing

Gear → **AUDIO SETUP**.



The Audio I/O window: device routing, output routing and the calibration reference

Device routing (upper half)

Control	What it does
Audio device type	The driver family: ASIO, Windows Audio, DirectSound (Windows); CoreAudio (macOS); ALSA/JACK (Linux). ASIO gives the lowest latency on Windows
Output	The interface that plays the processed sound
Input	The interface your guitar is connected to
Active channels	Tick the input channel the guitar uses and the output pair your monitors use
Sample rate	44.1 kHz and up. Higher rates cost more CPU; NAM captures are resampled internally as needed
Audio buffer size	The latency/stability trade-off. 128 or 256 samples is a good starting point
Control panel / Test	Opens the manufacturer's own panel (ASIO) and plays a test tone

Output routing (lower left). Four independent taps, each assigned to a physical output pair or switched **Off**:

Route	Signal it carries	Typical use
MAIN / POST MASTER	The finished player output	Monitoring and recording the final sound
PRE / NEUTRAL NAM	BLOCK NAM 2 before the player's tone stack, effects and cabinet	Re-amping, A/B comparison
DI / CLEAN INPUT	The untouched interface input	A clean safety track
WET / POST CAB	The processed post-cabinet branch	A separate processed recording path

A pair can only be used by one route: choosing a pair that is already taken resets that selector to **Off**. MAIN normally uses outputs 1/2. Leave the routes you do not need switched off so nothing is duplicated.

Interface input reference / 0 dBFS peak (lower right). A slider from 0 to 30 dBu, shown in dBu. It tells Input Cali what your interface's maximum input level is. It has no effect while Input Cali is off.

Everything in this window - device, channels, rate, buffer, reference and the four routes - is saved per machine and restored on the next launch. It is deliberately **not** stored in .bnpp presets, so a preset can travel between computers.

6. Signal flow

Each path processes its own audio in this order:

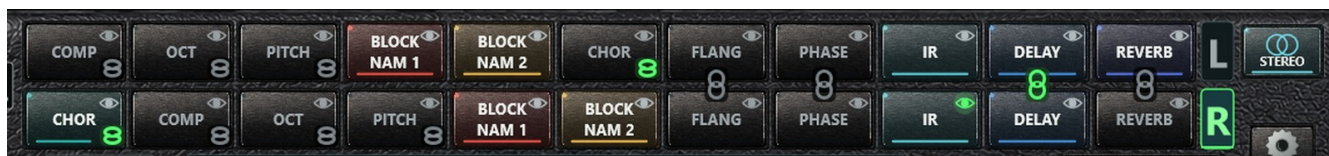
```

interface input
→ tuner tap (always the untouched DI, before everything)
→ INPUT GAIN (+ Input Cali offset)
→ GATE
→ [ rack blocks, in the order shown on screen ]
...effects before BLOCK NAM 2...
BLOCK NAM 1 (its own BASS/MID/TREBLE/PRESENCE)
BLOCK NAM 2 (then the main BASS/MID/TREBLE/PRESENCE)
...effects between BLOCK NAM 2 and IR...
IR (cabinet convolution → IR VOL → IR BLEND)
...effects after IR...
→ MASTER VOL
→ OUTPUT GAIN
→ POWER
→ MAIN / PRE / DI / WET output routes

```

BLOCK NAM 2 and IR are **anchors**: they cannot be dragged, and no block can be moved so that IR ends up before BLOCK NAM 2. Everything else is free.

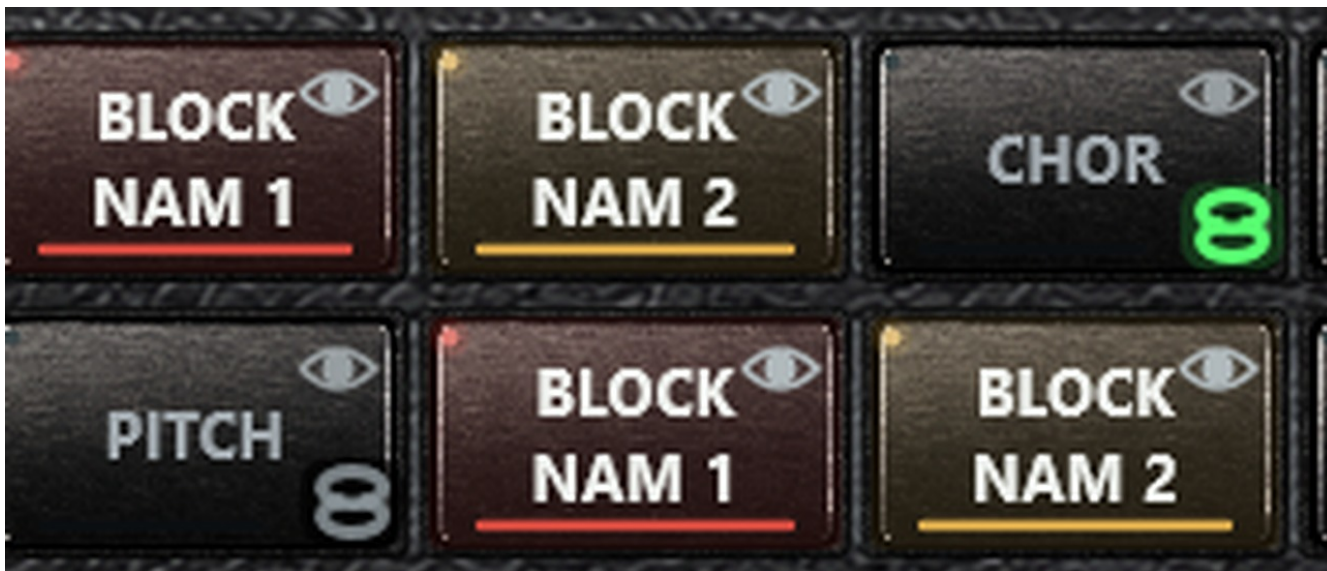
7. The rack: blocks, order and the eye



The two rack rows in STEREO: BLOCK NAM 1 (red), BLOCK NAM 2 (gold), an enabled CHOR, and the chain icons that link L and R

Eleven blocks per path, shown left to right in processing order. The default order is COMP · OCT · PITCH · BLOCK NAM 1 · BLOCK NAM 2 · CHOR · FLANG · PHASE · IR · DELAY · REVERB.

Reading a block



The parts of a block: name, eye, LED, colour bar and link icon

Part	Meaning
Body colour	Lit and tinted = enabled. Flat grey = bypassed
Small LED, top-left	Repeats the enabled state
Colour bar, bottom	The block's identity colour, lit while enabled
Eye, top-right	Chooses which block the shared controls and browsers describe. A viewed block has a green eye
Chain, bottom-right	Only in STEREO, only on effects: the L/R link. See chapter 13

What clicking does

Action	Result
Single click on the body	Enables or bypasses the block
Double-click on the body	Effects: opens the editor. BLOCK NAM 1 / BLOCK NAM 2: opens the model chooser. IR: opens the IR chooser
Click on the eye	Makes that block the one the shared controls edit. A bypassed block cannot be viewed
Drag horizontally	Moves the block; the target position is outlined in white
Drop a file on a block	Loads it: .nam on a NAM block, .wav on the IR block; a folder loads its first valid file
Hover BLOCK NAM 1 / BLOCK NAM 2	Shows the capture information card

A NAM or IR block can only be enabled once it holds a valid file. Clicking an empty one while its file is still loading means "switch on when ready", not "invert".

8. BLOCK NAM 1 and BLOCK NAM 2

Both blocks run any compatible .nam capture; the player does not assume what was captured. BLOCK NAM 1 sits earlier in the chain and is the natural place for a pedal or preamp capture, BLOCK NAM 2 for an amplifier - but nothing stops you using them the other way round.



The two browsers: NAM models on the left with its four selectors, IR files on the right, IR VOL between them

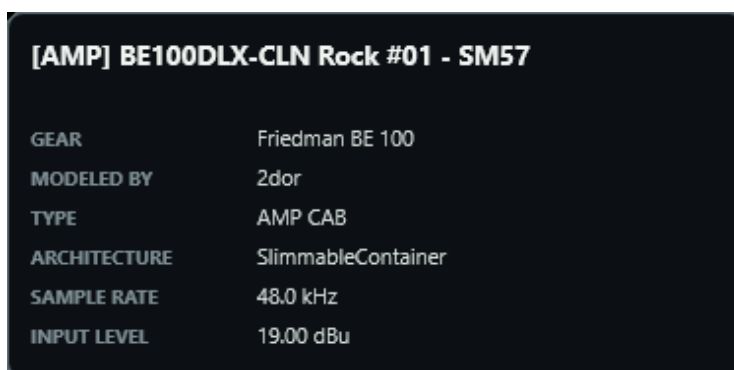
Loading a capture

Route	How
BROWSE LOCAL	Pick one .nam, or a folder. A folder adds every capture it contains to the list and selects the first. Tick DEEP SEARCH in the file dialog to include subfolders
The list	Click any row to load it; the ▲ / ▼ buttons step through the list and grey out at the ends
TONE3000	Opens the online library inside the player - see chapter 16
Drag and drop	Drop a .nam or a folder on the list or directly on the block
Double-click the block	Opens the file chooser for that block

Each block remembers its own folder, its own deep-search setting, its own list and its own selection, per path. Selecting a capture for BLOCK NAM 2 also resets that path's controls to safe defaults and switches the block on; the cabinet selection is left alone.

Architecture is automatic. The player identifies NAM A1, A2 Standard and A2 Nano and dispatches A2 models through NAM Core's fast path. There is no manual switch. If a file is not a valid single-input/single-output NAM, the load is refused and the block stays as it was.

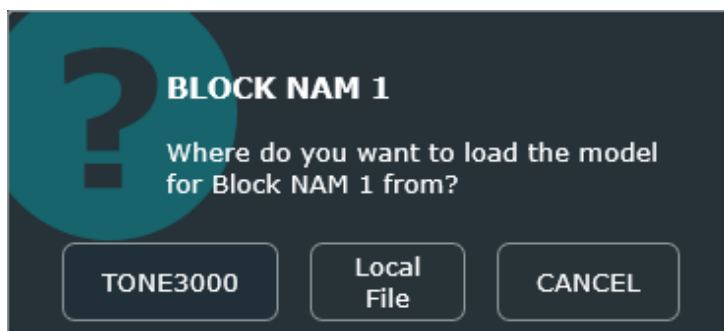
The capture information card. Hovering a NAM block or any row of the NAM list shows the metadata written into the .nam header - title, gear make and model, who modelled it, gear type, architecture, sample rate and input reference level - plus artwork when available.



The capture information card

Artwork is any image sitting beside the capture with the same base name (My Capture.nam → My Capture.png, .jpg, .jpeg or .webp). TONE3000 downloads save it automatically; for your own captures, drop an image next to the file. Only the header of the file is read, so hovering a long list costs nothing.

Loading BLOCK NAM 1 from the front panel. The rack's + **BLOCK NAM 1** button (and the NAM 1 tab) asks where the capture should come from:



Choosing the source for BLOCK NAM 1

9. Cabinet impulse responses

BROWSE IR loads a .wav response, or a folder: the player keeps the short WAV files that are usable as impulse responses, lists them and selects the first. If nothing suitable is found it reports **IR not found**.

- Mono and stereo responses are accepted; a stereo file is summed to mono.
- Up to 8192 samples are used, resampled to the current device rate.
- **The original gain is preserved - IRs are never normalized.**
- Click the selected row a second time to deselect it: the response is cleared

and the IR block is bypassed.

- **DELETE** removes the selected response from the IR folder, after asking.
- Loading or changing a response crossfades, so it never clicks.

Two controls shape the cabinet:



IR VOL, with its own scale and the (L)/(R) caption in STEREO

- **IR BLEND** mixes the signal without cabinet convolution against the convolved signal, 0 - 100 %.
- **IR VOL** (the vertical fader) sets the level of the cabinet branch alone, **-24 dB to 0 dB**, default **-12 dB** to leave headroom for hot captures. The current value is printed under the scale.

10. Amplifier controls, knob by knob

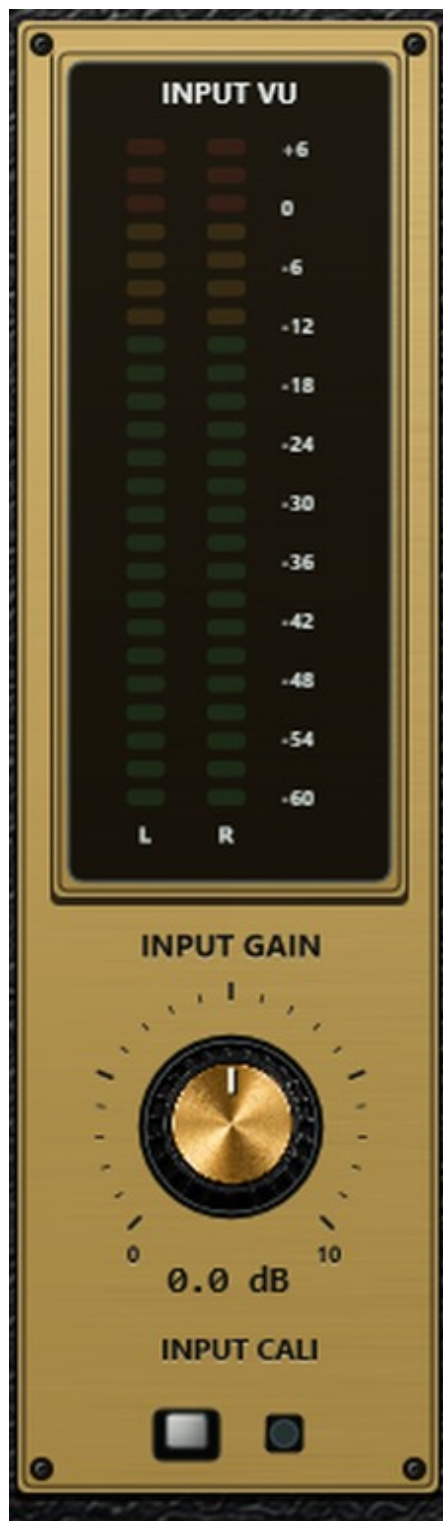


The main control row and the NAM/IR selectors above it

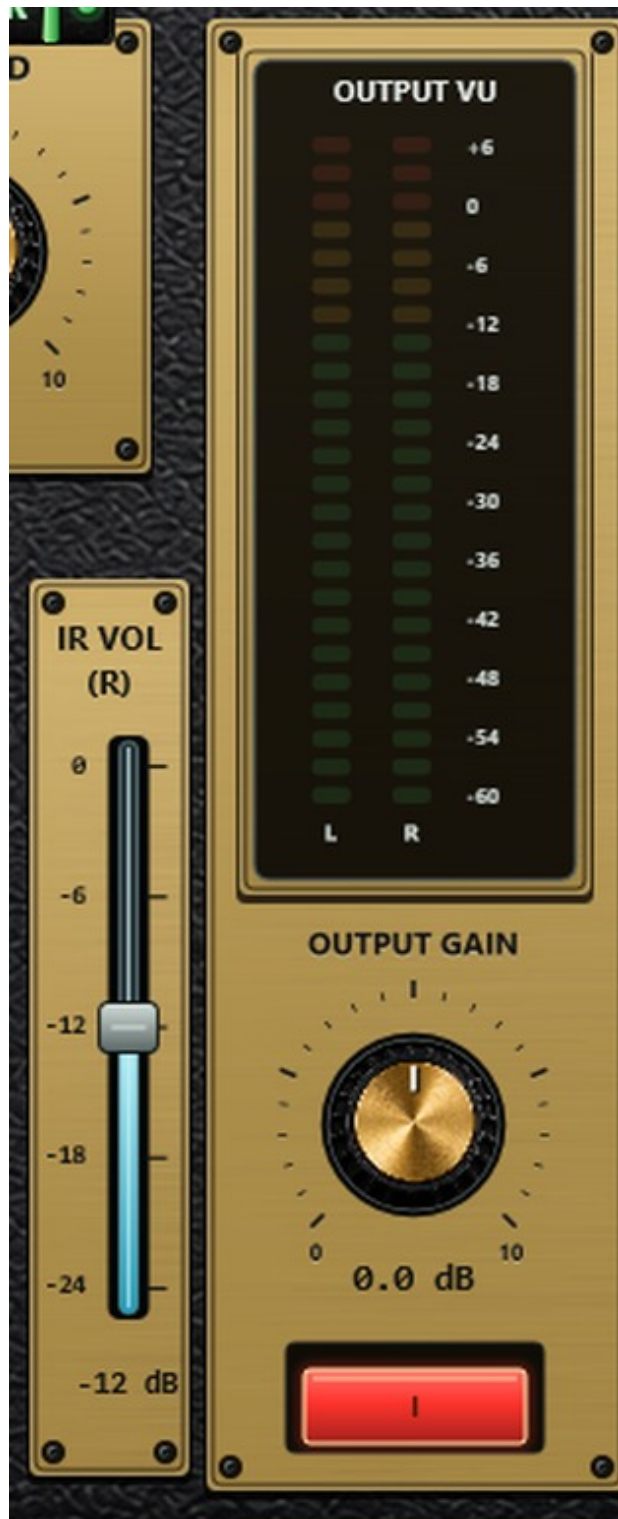
Operating a knob: drag vertically or horizontally; the exact value is printed underneath. **Double-click restores the default** - 12 o'clock for everything except GATE, which returns to fully left (OFF).

Control	Range	Default	Notes
INPUT GAIN	-12 ... +12 dB	0.0 dB	Level into the NAM chain; Input Cali is added on top
GATE	OFF ... -80 ... 0 dB threshold	OFF	At the far left it is a true bypass. Attack 1.5 ms, hold 35 ms, release 90 ms, 3 dB hysteresis
BASS	±12 dB @ 70 Hz	0.0 dB	Peaking, Q 0.72
MID	±12 dB @ 750 Hz	0.0 dB	Peaking, Q 0.72
TREBLE	±12 dB @ 4 kHz	0.0 dB	Peaking, Q 0.72
PRESENCE	±12 dB @ 6 kHz	0.0 dB	Peaking, Q 0.72
MASTER VOL	-12 ... +12 dB	0.0 dB	After the chain, before Output Gain
IR BLEND	0 ... 100 %	50 %	Dry against cabinet
IR VOL	-24 ... 0 dB	-12 dB	Cabinet branch only
OUTPUT GAIN	-12 ... +12 dB	0.0 dB	Final level
POWER	On / off	On	Mutes the processed path; the switch glows red while on
INPUT CALI	On / off	Off	See chapter 15

The tone stack follows the selected NAM block. With a **NAM 1** selector active, BASS / MID / TREBLE / PRESENCE drive BLOCK NAM 1's own four filters; with a **NAM 2** selector active they drive the main tone stack. Each set keeps its own values, and switching selectors recalls them. In the plug-in and in STEREO the plate prints which one you are editing.



Input column: Input VU, INPUT GAIN and INPUT CALI with its LED



Output column: Output VU, OUTPUT GAIN and POWER

11. The eight effects in full

Every effect offers **three algorithms** and **six parameters**. Double-click a block to open its editor; every knob shows a real unit, and double-clicking a knob restores its default.



The effect editor, here for DELAY

The **ACTIVE** switch at the top right of the editor is the same bypass as clicking the block. **TYPE** chooses the algorithm. Parameter, algorithm and bypass changes are smoothed, delay-family reads are interpolated and feedback is bounded, so nothing clicks or runs away.

COMP - compressor

Algorithms: **Studio VCA, Optical, FET Punch.**

Parameter	Range	Default
Threshold	-55 ... -2 dB	-31.2 dB
Ratio	1 ... 20 :1	7.7:1
Attack	1 ... 100 ms	15.9 ms
Release	20 ... 600 ms	223 ms
Makeup	-12 ... +12 dB	0.0 dB
Mix	0 ... 100 %	100 %

DELAY

Algorithms: **Digital Studio, Tape Echo, Analog BBD.**

Parameter	Range	Default
Time	20 ... 1200 ms	398 ms
Feedback	0 ... 92 %	32 %
Mix	0 ... 100 %	28 %
Tone	0 ... 100 %	65 %
Mod	0 ... 100 %	8 %
Level	-12 ... +12 dB	0.0 dB

CHOR - chorus

Algorithms: **Studio, Ensemble, Tri-Chorus.**

Parameter	Range	Default
Rate	0.05 ... 5 Hz	1.24 Hz
Depth	0.5 ... 20 ms	10.3 ms
Mix	0 ... 100 %	35 %
Delay	4 ... 30 ms	11.8 ms
Feedback	-65 ... +65 %	0 %
Level	-12 ... +12 dB	0.0 dB

FLANG - flanger

Algorithms: **Analog, Through-Zero, Jet.**

Parameter	Range	Default
Rate	0.03 ... 2.5 Hz	0.57 Hz
Depth	0.1 ... 9 ms	5.0 ms
Mix	0 ... 100 %	35 %
Feedback	-85 ... +85 %	20 %
Manual	0.2 ... 5 ms	1.4 ms
Level	-12 ... +12 dB	0.0 dB

PHASE - phaser

Algorithms: **4 Stage, 8 Stage, 12 Stage.**

Parameter	Range	Default
Rate	0.03 ... 4 Hz	0.82 Hz
Depth	0 ... 100 %	70 %
Mix	0 ... 100 %	40 %
Feedback	-75 ... +75 %	12 %
Centre	180 ... 2200 Hz	887 Hz
Level	-12 ... +12 dB	0.0 dB

REVERB

Algorithms: **Studio Room, Plate, Concert Hall.**

Parameter	Range	Default
Size	0 ... 100 %	55 %
Damping	0 ... 100 %	50 %
Mix	0 ... 100 %	25 %
Width	0 ... 100 %	80 %

Parameter	Range	Default
Freeze	OFF / ON	OFF
Level	-12 ... +12 dB	0.0 dB

OCT - octaver

Algorithms: **Poly Clean**, **Classic Mono**, **Organ**.

Parameter	Range	Default
Oct Down	0 ... 100 %	45 %
Oct Up	0 ... 100 %	0 %
Dry	0 ... 100 %	80 %
Tone	0 ... 100 %	50 %
Tracking	0 ... 100 %	65 %
Level	-12 ... +12 dB	0.0 dB

PITCH - pitch shifter

Algorithms: **Studio**, **Low Latency**, **Vintage**.

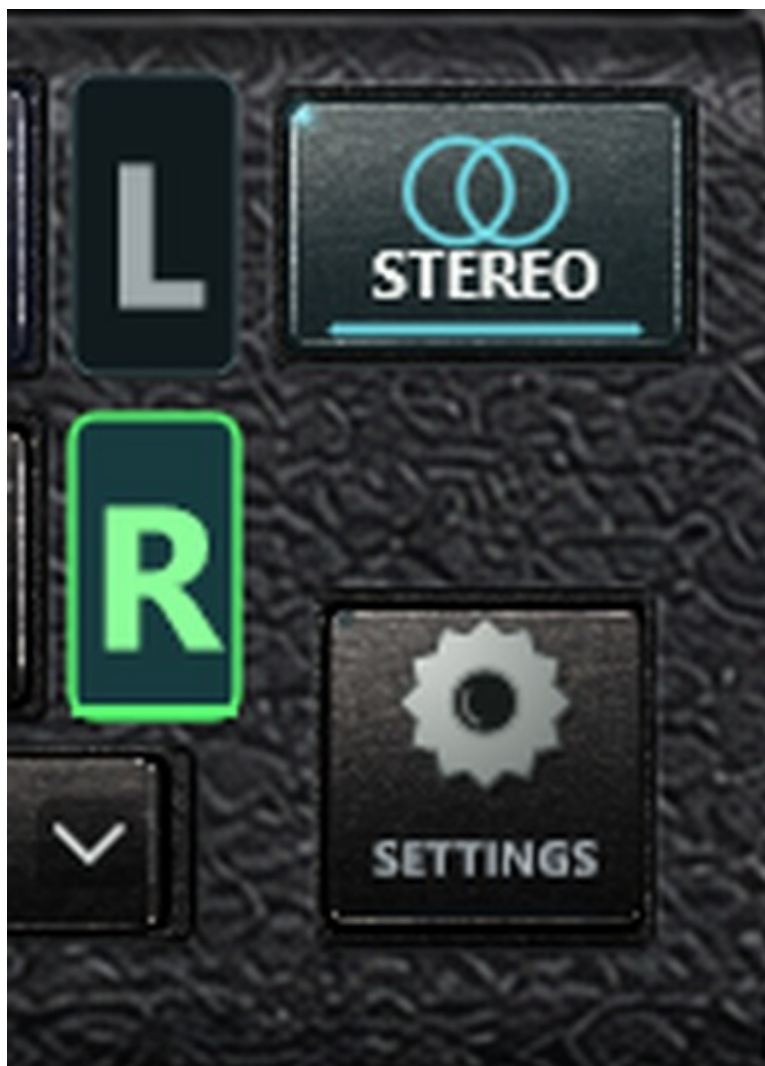
Parameter	Range	Default
Semitones	-12 ... +12 st	0.0 st
Mix	0 ... 100 %	100 %
Window	20 ... 120 ms	65 ms
Feedback	0 ... 55 %	0 %
Fine	-100 ... +100 ct	0 ct
Level	-12 ... +12 dB	0.0 dB

12. MONO and STEREO: two complete rigs



STEREO with SPLIT L/R: two rack rows, four NAM selectors, two IR selectors and stereo meters

MONO / STEREO (top right of the rack) switches between one and two audible paths.



The MONO / STEREO switch and the settings gear

- **MONO**: the interface input is summed into the left chain, and its result feeds both channels of every routed output pair. Right-hand selectors, the second rack row and the input-mode switch are hidden.

- **STEREO**: both chains run. Two rack rows appear, the meters split into L and R, and the L / R tabs beside the racks choose which path the shared controls edit.

DUAL MONO / SPLIT L/R (beside the preset bay, STEREO only):

Position	Input handling
DUAL MONO	The summed interface input feeds both chains - one guitar through two independent rigs
SPLIT L/R	Input channel 1 feeds the left chain, input channel 2 feeds the right chain

This setting belongs to the application state, not to presets.

The selectors.



NAM 1 L · NAM 2 L · NAM 1 R · NAM 2 R and IR L · IR R

Selector	Chooses
NAM 1 L NAM 2 L NAM 1 R NAM 2 R	The NAM block the tone controls and the NAM browser act on
IR L IR R	The cabinet that IR BLEND and IR VOL act on
L / R tabs	The path every other shared control acts on

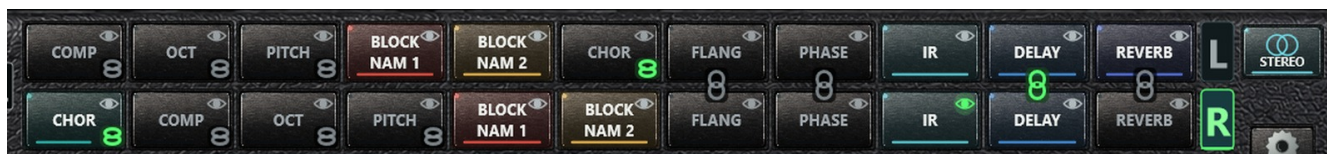
Which control follows what

Control	Follows
INPUT GAIN, GATE, MASTER VOL, OUTPUT GAIN, POWER, INPUT CALI, TUNER	the selected path
BASS, MID, TREBLE, PRESENCE	the selected NAM block of that path (NAM 1 has its own set)
IR BLEND, IR VOL	the selected cabinet
Tuning reference, skin, Audio Setup, routing, updates	the whole application

In MONO only the two left selectors are shown, and the IR VOL legend drops its (L) / (R) suffix.

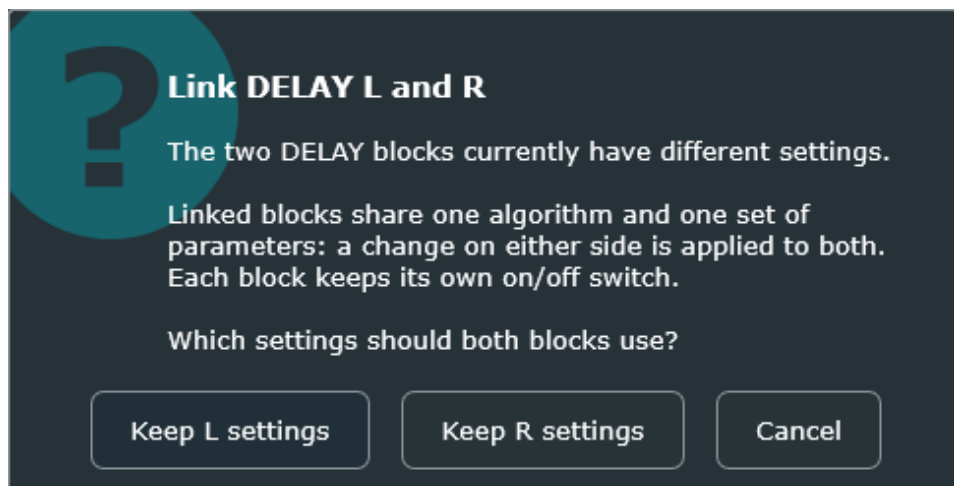
13. Linking effects between L and R

In STEREO you can lock an effect of the left chain to the same effect of the right chain, so you only have to dial it once.



Linked pairs are green; unlinked pairs are grey. A pair that no longer shares a column shows its icon in the corner of both blocks

- **Where the icon is.** When both blocks are in the same column, one chain icon sits on the seam between the two rack rows and joins them. If you move one of them so the pair no longer shares a column, the icon moves to the **lower-right corner of each block** - it still works exactly the same.
- **Colour.** Green = linked. Grey = not linked.
- **Click to link or unlink.**
- **What is shared:** the algorithm and the six parameters. Editing either block changes both.
- **What is not shared:** the on/off switch and the position in the chain. The two blocks can be both on, both off, or one of each - linking never touches them.
- **Linking two blocks that differ** asks which side to keep:



Choosing which side's settings to keep

- **Unlinking** leaves both blocks exactly as they sound at that moment; from then on they are edited separately.

- **BLOCK NAM 1, BLOCK NAM 2 and IR cannot be linked.**
- Links are saved with the session and inside two-path .bnpp presets.

14. Tuner



The tuner row: the TUNER button, the LED display and the tuning selector

Press **TUNER**; the lettering lights green and the button looks pressed. The display shows the detected note with a left/right LED indication for flat, centred or sharp.

- The tuner listens to the **untouched DI**, before the gate, the NAM blocks and the effects, and is not part of the draggable order.

- Range roughly **65 - 700 Hz**, at every sample rate from 44.1 to 192 kHz.
- Within **±5 cents** the display reads centred and turns green; beyond about 40 cents it turns red.

- **TUNING** offers **STANDARD, DROP D, D STANDARD, Eb** and **OPEN G**.

- Play one isolated string at a useful level and let the previous note decay.

15. Input Cali (automatic input calibration)

NAM captures are trained at a known input level. Input Cali matches your interface to that level so a capture sounds the way it was captured.

With Input Cali on, the player computes:

$$\text{calibration gain (dB)} = \text{interface input reference (dBu)} - \text{NAM input reference (dBu)}$$

- The interface value is the slider in Audio Setup.
- The NAM value comes from the capture's `input_level_dbu` metadata.

- When a capture has no metadata, the standard NAM reference of **+12 dBu** is assumed.

- The result is clamped to **-24 ... +24 dB** and applied before the NAM blocks.

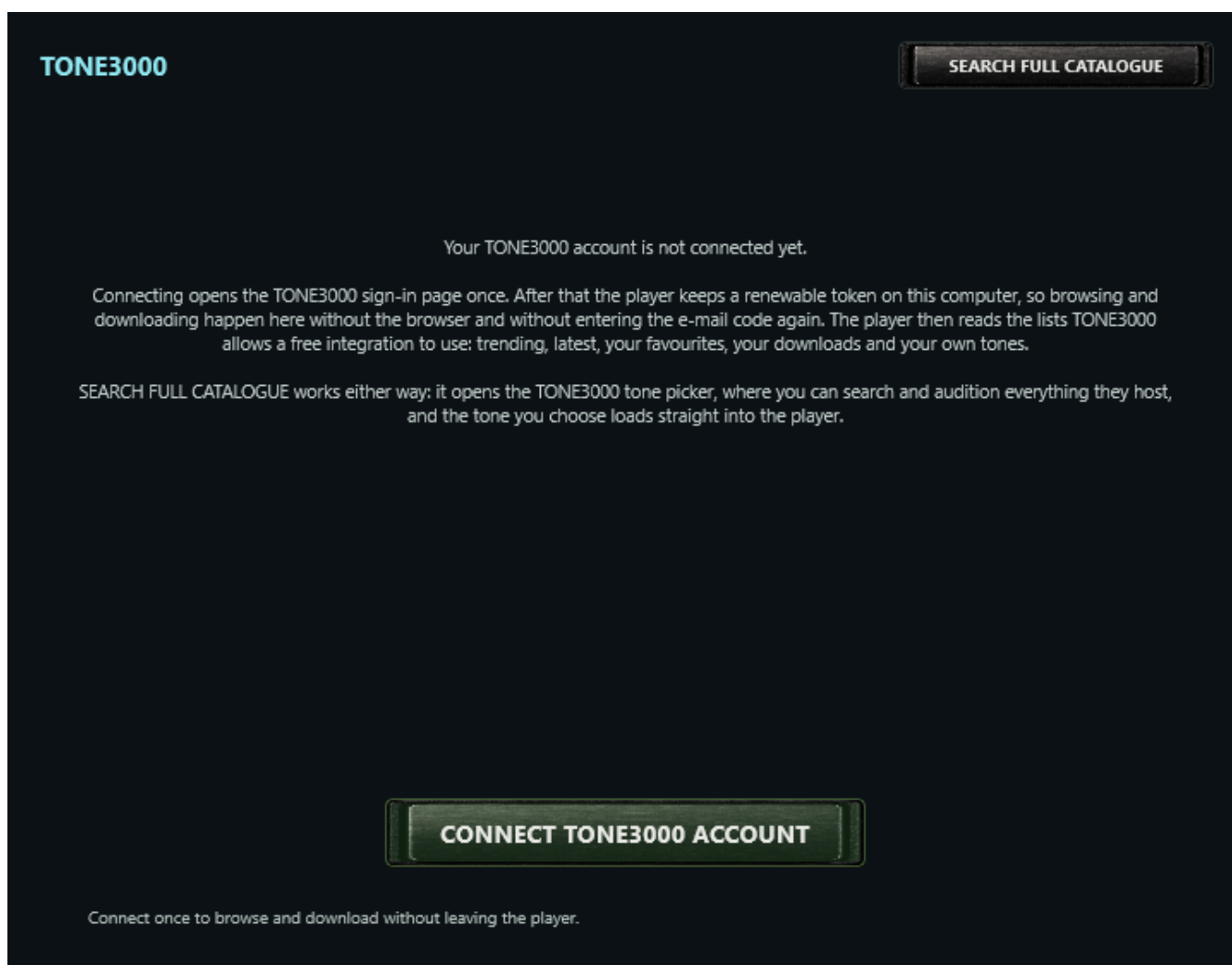
- The small LED beside INPUT CALI is green while it is active.

- While a replacement capture is still loading, the gain of the running capture is held - it never jumps to 0 dB mid-note.

Input Cali changes gain only. It never rewrites or normalizes a model. If your interface has several input modes (instrument / line / pad), enter the dBu value of the mode you are actually using. For a PreSonus Studio 24c, that is +10 dBu.

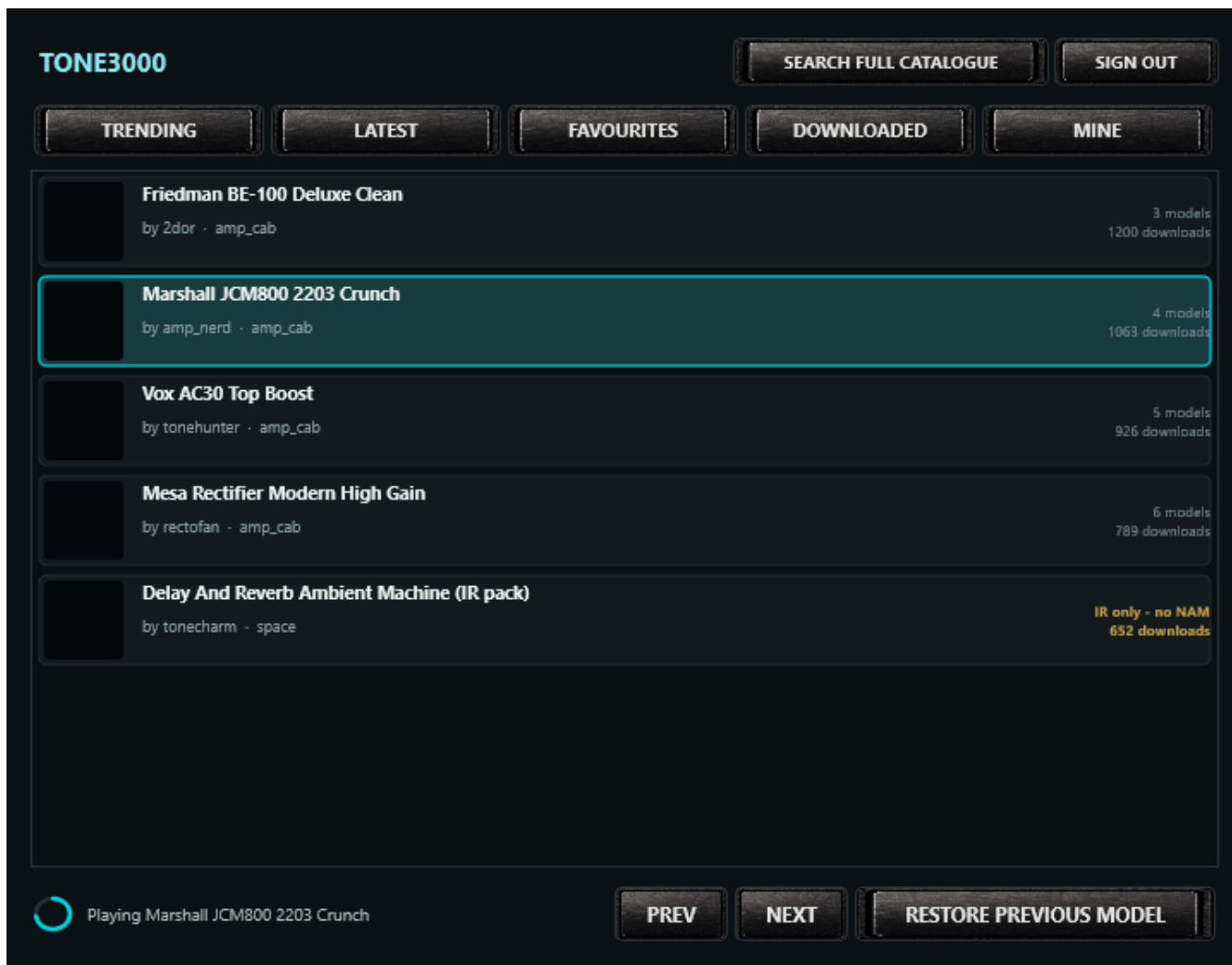
16. TONE3000 inside the player

TONE3000 (green button above the NAM list) opens the online capture library in a window of the player.



Before connecting an account

Connecting once. The first use asks to connect your account. TONE3000 signs in with an e-mail link, so that opens their sign-in page once; the player then keeps the returned refresh token, encrypted with a key derived from this machine, in `tone3000.session` beside the application. Every later listing and download renews the token silently, so no browser is needed again on that computer. **SIGN OUT** deletes the file.



The library: five lists, artwork, author, model count and paging

Browsing and auditioning.

Element	What it does
TRENDING · LATEST · FAVOURITES · DOWNLOADED · MINE	The five listings a free-tier integration may read
Selecting a row	Loads the capture into the current NAM block immediately. The first capture of the tone is fetched and plays as soon as it is on disk; the rest of the tone and its artwork keep downloading in the background
Selecting another row	Cancels whatever is still downloading and auditions the new one instead
A tone you already downloaded	Loads instantly from your downloads folder
Rows marked IR only - no NAM	Contain only impulse responses; there is nothing to load into a NAM block, so they are not auditioned
RESTORE PREVIOUS MODEL	Puts back the capture the block held before this window was opened . Disabled if the block was empty
Closing the window	Keeps the last capture you auditioned
PREV / NEXT	Paging
The turning indicator and the status line	Show progress: *Downloading tone (n / m)*, then *Playing <capture>*
SEARCH FULL CATALOGUE	Opens TONE3000's own picker, with search and auditioning, inside the player. When you choose a tone the picker closes by itself and the download starts
SIGN OUT	Forgets the account on this computer

Downloads are saved under your TONE3000 downloads folder, one folder per tone, with the tone artwork beside each capture so the information card can show it.

WebView2 (Windows only). The embedded sign-in and the catalogue picker use Microsoft WebView2, which ships with Windows 11 and with Edge on Windows 10. If it is missing or unavailable the player offers an official Microsoft download, your external browser, or cancelling - **nothing is ever installed automatically**, and no Microsoft installer is included in the package. Local audio and local files never need WebView2. macOS uses WKWebView; Linux falls back to the system browser. External and embedded browsers do not share cookies, so a session created in an older build may need one sign-in inside the player.

17. Portable .bnpp presets

SAVE PRESET writes one self-contained archive holding **both** paths:

- the BLOCK NAM 2 capture of each path;
- the BLOCK NAM 1 capture of each path, when loaded;
- the selected IR of each path, when loaded (up to six embedded resources);
- rack order, algorithms, all effect parameters and every bypass state;
- all knob, switch and tuner settings;
- the L/R effect links.

Embedded resources are verified with SHA-256 when loaded, extracted to the application's preset cache and restored. Audio device, interface reference and physical routing are deliberately excluded so a preset can move between computers.

LOAD PRESET restores it. A preset written by the single-path player, or by NAM PLAYER DUAL, loads into the **currently selected path**. A preset written here still opens in the older single-path player, because the left path is also written in the old flat layout.

A capture must be loaded in BLOCK NAM 2 before a .bnpp can be saved.

18. What is remembered, and where

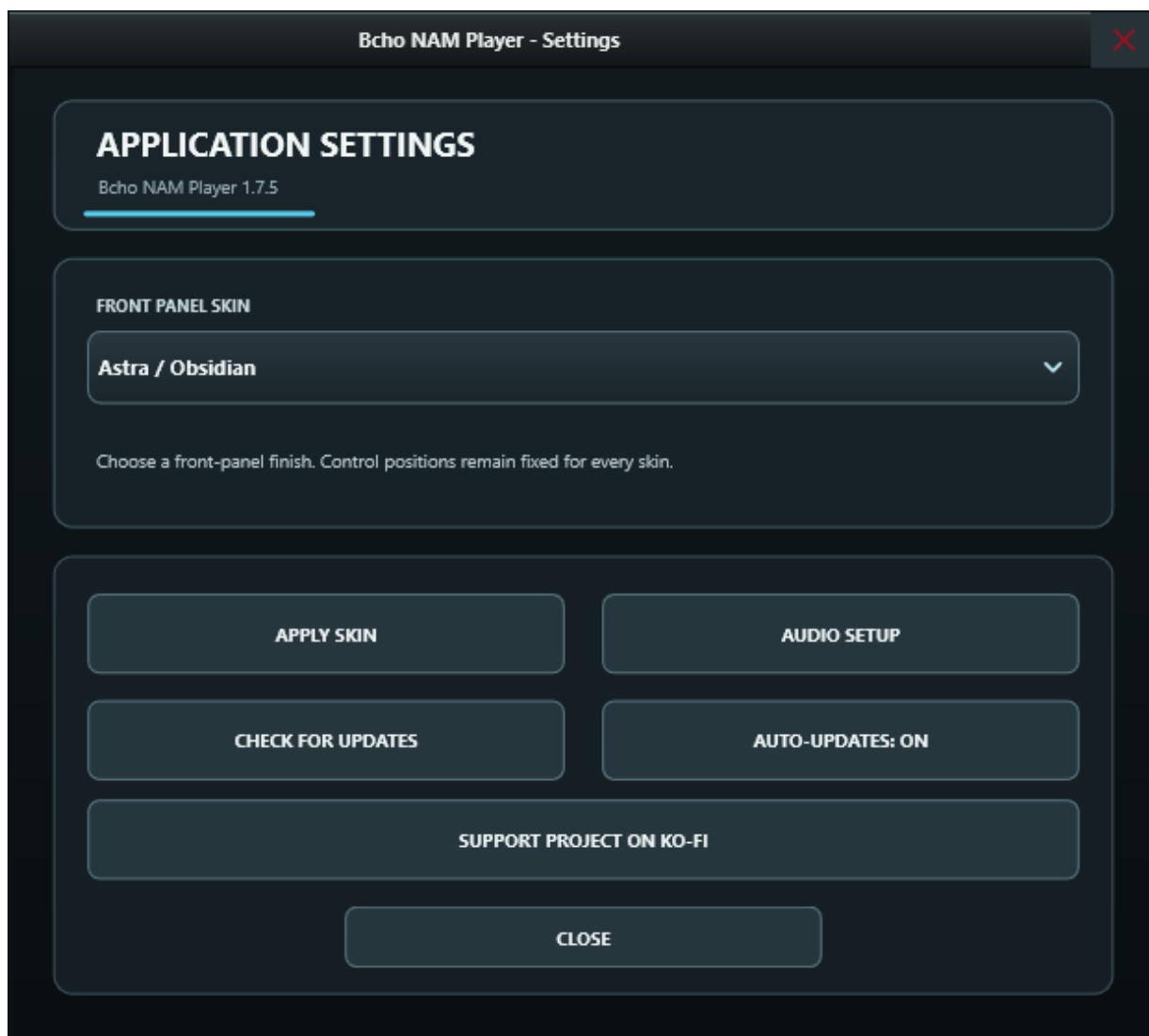
Session state is written on a normal exit and restored on the next launch: both paths, all controls, the selected captures and IRs, block order and bypass states, links, MONO/STEREO, DUAL MONO/SPLIT, skin, audio device, channels, rate, buffer, interface reference and the four routes.

File	Where	What it holds
NAM_A2_HEAD.xml	Beside the application when that folder is writable, otherwise the per-user data folder	The session
tone3000.session	Same folder	The encrypted TONE3000 refresh token
Presets / PresetCache	Same folder	Presets you save and the resources extracted from loaded ones
Per-user data folder	Windows %APPDATA%\Bcho\BchoNAMPlayerDual, macOS ~/Library/Application Support/Bcho/..., Linux ~/.config/Bcho/...	Used when the application folder is read-only (macOS bundles, Applimages)

NAM and IR files are referenced by path, not copied into the session: keep them where they were when you saved. Presets, by contrast, embed them.

19. Settings, skins, updates and support

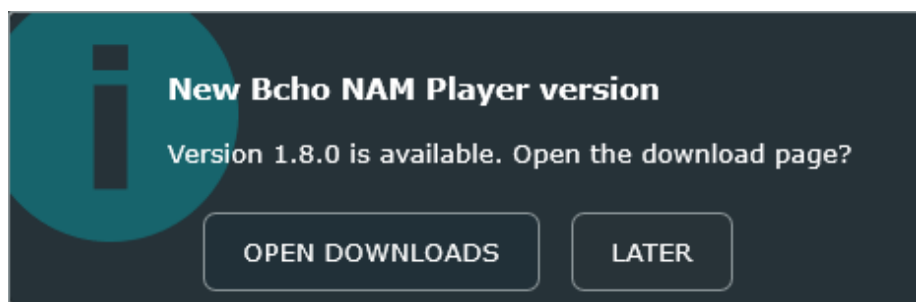
Press the gear to the right of the rack.



Application settings

Control	What it does
FRONT PANEL SKIN	Previews a finish immediately
APPLY SKIN	Makes the previewed finish permanent. Closing without applying restores the previous one
AUDIO SETUP	Opens Audio Setup
CHECK FOR UPDATES	Compares your version with the latest public release
AUTO-UPDATES: ON / OFF	Whether that check also runs at startup
SUPPORT PROJECT ON KO-FI	Opens ko-fi.com/bchosoft in your browser
CLOSE	Closes the window

The eleven finishes: Astra / Obsidian · Tribal / Etched Titanium · Skulls / Bone & Carbon · Hippie / Sunset Paisley · Graffiti / Electric Ink · Purple Velvet / Amethyst · Stainless Steel / Precision · Ripped Black Denim / Roadworn · Blue Denim / Indigo · Spiderwebs / Black Widow · Classic Black / Levant Tolex. Every finish uses identical geometry, so no control ever moves; only materials, knob designs and legend contrast change. The plug-in always uses Astra / Obsidian.



The update notice

Update checking only reports availability and opens the download page if you ask it to. It never replaces files, models or presets.

20. Meters and animated cones

INPUT VU shows the incoming level and **OUTPUT VU** the final processed level, from +6 down to -60 dB. In STEREO each meter splits into an L and an R column.



The cabinet: both cones follow the real output level

Both cones react to the real final output RMS: louder playing, or a higher MASTER VOL / OUTPUT GAIN, gives a larger excursion; silence and POWER off return them smoothly to rest. Only the cone surfaces move - rings, screws, grille, cabinet and logo stay fixed. The animation runs at 60 frames per second on the interface thread and never touches, delays or feeds back into the audio.

21. Mouse and keyboard reference

Where	Action	Result
Any knob	Drag up/down or left/right	Change the value
Any knob	Double-click	Restore the default (GATE returns to OFF)
Rack block	Single click	Enable / bypass
Rack block	Double-click	Open its editor or file chooser
Rack block	Drag sideways	Reorder (BLOCK NAM 2 and IR are fixed)
Rack block	Drop a file or folder	Load it into that block
Eye icon	Click	View that block with the shared controls
Chain icon	Click	Link / unlink that effect between L and R
NAM or IR list	Click a row	Load it
IR list	Click the selected row again	Deselect: clears the response and bypasses the IR block
NAM or IR list	Drop a file or folder	Load it
▲ / ▼	Click	Previous / next entry in the list

Where	Action	Result
Dialog window	Esc	Close
Any control	Hover	Tooltip

22. Troubleshooting

Symptom	What to check
No sound at all	POWER is on; MAIN route points at the outputs you monitor; the right input channel is ticked in Audio Setup; INPUT VU moves when you play
Sound, but no amp tone	A capture is loaded and BLOCK NAM 2 is lit; the chain is not entirely bypassed (a bypassed chain passes dry audio)
A NAM block will not switch on	Load a valid capture into that block and wait for the load to finish; a valid capture enables it automatically
IR block stays bypassed	Load a valid response, and check you have not clicked the selected row a second time (that deselects it)
Clicks or dropouts while playing	Raise the audio buffer, use the manufacturer ASIO driver on Windows, and avoid very small buffers with four NAM blocks running
A folder appears empty	It must contain .nam files, or short valid .wav responses, directly inside the folder - unless DEEP SEARCH is ticked
Level does not match other NAM software	Check the interface reference in Audio Setup and whether Input Cali is on
A capture sounds too loud or too quiet	Check IR VOL (-12 dB by default) and Input Cali before changing INPUT GAIN
The session did not come back	The player saves on a normal exit; after a forced quit the previous session is restored instead
Models or IRs went missing	They are referenced by path. Put them back, or load a .bnpp preset, which embeds them
TONE3000 list is empty	Check the account, the network and - on Windows - WebView2; or use the external browser
macOS refuses the first launch	Right-click the application and choose Open
Linux AppImage will not start	chmod +x it and confirm the system has a working x86_64 audio stack

23. Technical specifications

Item	Value
Signal paths	2 independent (L / R)
NAM blocks	2 per path (4 in total)
Cabinet IR	1 per path
Effects	8 per path, 3 algorithms and 6 parameters each
Rack positions	11 per path, freely reorderable around the BLOCK NAM 2 and IR anchors
NAM architectures	A1, A2 Standard, A2 Nano - detected automatically
Sample rates	Whatever the device offers, 44.1 - 192 kHz
Buffer size	Whatever the driver offers; the engine internally uses at least 32 samples and splits oversized host buffers
IR length	Up to 8192 samples, resampled to the device rate, never normalized
Tuner range	~65 - 700 Hz, ±5 cents centred
Auto Cal range	-24 ... +24 dB
Start-up fade	60 ms from silence

Item	Value
NAM change fade	15 ms out, silence until the new capture is installed, 20 ms in (raised-cosine)
IR change fade	12 ms crossfade
Native window	1537 x 1023, responsive, minimum 50 %
Interface refresh	60 fps, entirely off the audio thread

24. Integrity, third-party code and licences

Compare downloaded ZIP hashes against SHA256SUMS.txt. Bcho NAM Player includes JUCE and Neural Amp Modeler Core; their notices are in THIRD_PARTY_NOTICES.md.

The application grants no rights over third-party NAM captures or impulse responses. Follow the licence supplied with each capture or response, including anything you download through TONE3000 or share inside a .bnpp preset.